

ORYOL, Russian Federation

WMO: 279060

Lat: 52.9318N

Lon: 36.0189E

Elev: 196

StdP: 98.99

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.4	-20.0	-26.6	0.3	-23.1	-23.2	0.5	-19.9	10.3	-3.0	9.6	-2.5	2.3	10	0.577

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	30.5	19.9	28.6	19.3	26.9	18.5	21.2	27.5	20.4	26.5	19.6	25.3	3.5	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.1	14.3	24.2	18.4	13.5	23.2	17.5	12.9	22.3	62.4	27.5	59.6	26.7	56.8	25.4	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.2	7.3	-26.5	32.7	3.2	2.2	-28.8	34.3	-30.6	35.6	-32.4	36.9	-34.7	38.5		
(5)			-26.7	22.7	3.1	1.0	-28.9	23.4	-30.7	23.9	-32.5	24.5	-34.7	25.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.7	-6.6	-6.2	-1.0	7.8	14.3	17.7	20.0	18.6	12.9	6.5	0.5	-4.4		
	DBStd	10.76	6.42	6.59	4.82	4.71	4.67	3.74	3.32	3.71	4.16	4.41	4.93	6.14		
	HDD10.0	2299	515	453	339	98	14	1	0	0	18	126	287	448		
	HDD18.3	4409	773	687	598	317	140	56	20	42	168	367	536	706		
	CDD10.0	1111	0	0	0	31	148	233	311	266	106	17	1	0		
	CDD18.3	181	0	0	0	0	16	38	73	49	6	0	0	0		
	CDH23.3	1592	0	0	0	6	148	305	584	491	58	1	0	0		
	CDH26.7	437	0	0	0	0	26	64	174	167	6	0	0	0		
(14) Wind	WSAvg	3.5	4.0	4.0	4.0	3.7	3.4	3.2	2.8	3.0	3.0	3.5	3.8	3.9		
(15) Precipitation	PrecAvg	640	49	39	39	44	48	68	83	57	63	59	47	44		
	PrecMax	843	95	75	88	91	89	142	171	155	184	106	85	134		
	PrecMin	433	24	16	9	2	15	14	25	3	5	7	3	12		
	PrecStd	90	17	15	21	23	19	32	36	45	36	30	22	26		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.5	4.5	14.1	24.0	29.2	30.9	33.5	34.5	27.6	20.6	12.9	7.7		
		MCWB	3.6	3.0	8.1	12.8	18.4	19.3	20.3	19.7	16.4	13.7	10.9	6.6		
	2%	DB	2.8	3.0	9.7	20.4	27.0	28.4	30.4	30.7	25.1	17.3	9.9	5.7		
		MCWB	1.9	2.0	5.0	11.7	16.8	19.1	20.4	19.9	16.1	12.9	8.6	4.9		
	5%	DB	1.7	2.1	7.0	18.2	24.8	26.7	28.5	28.1	22.4	14.7	8.3	3.2		
		MCWB	0.9	1.1	3.6	10.9	16.2	18.4	20.1	19.2	15.3	11.6	7.2	2.5		
	10%	DB	0.8	1.4	5.0	15.8	22.6	24.9	26.7	25.9	19.8	12.6	6.8	2.0		
		MCWB	0.1	0.6	2.6	9.7	15.1	17.7	19.4	18.0	14.4	10.4	5.8	1.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.7	3.2	8.1	14.4	19.6	21.5	22.7	22.1	18.3	15.2	11.1	6.9		
		MCDB	4.1	3.9	13.3	20.4	27.6	27.4	29.1	28.2	24.3	18.2	12.7	7.5		
	2%	WB	2.1	2.2	5.7	12.6	18.2	20.3	21.6	20.8	17.1	13.4	8.9	4.8		
		MCDB	2.6	2.7	8.9	17.9	24.7	26.3	27.9	27.6	22.7	16.5	9.7	5.5		
	5%	WB	1.0	1.3	4.1	11.6	17.0	19.3	20.8	19.9	16.0	12.0	7.4	2.7		
		MCDB	1.5	1.9	6.6	16.9	23.0	24.9	26.6	26.9	20.4	14.4	8.1	3.2		
	10%	WB	0.2	0.7	2.8	10.5	15.9	18.4	20.0	18.9	15.0	10.7	5.8	1.5		
		MCDB	0.8	1.3	4.5	15.2	21.4	23.5	25.2	24.7	19.0	12.4	6.7	1.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.9	5.8	6.6	9.2	10.2	9.8	9.7	10.7	9.3	6.9	4.2	4.4		
		MCDBR	3.3	4.0	9.2	12.7	12.8	12.1	12.3	13.8	13.4	10.7	5.8	4.0		
	5% WB	MCWBR	3.3	3.6	5.6	6.2	5.5	5.2	4.8	5.2	6.1	6.5	4.8	3.9		
		MCWBR	3.2	3.9	8.1	11.0	11.1	10.5	10.4	12.4	11.4	9.4	5.0	4.0		
(40) Clear-Sky Solar Irradiance	taub		0.296	0.332	0.355	0.430	0.396	0.385	0.419	0.430	0.390	0.357	0.332	0.300		
		taud	2.126	2.043	2.097	2.097	2.297	2.348	2.269	2.223	2.304	2.356	2.249	2.152		
	Ebn at Noon		655	740	815	797	854	867	826	791	778	726	614	572		
		Edh at Noon	71	106	124	142	122	117	125	124	102	79	64	59		
(44) All-Sky Solar Radiation	RadAvg		0.84	1.83	3.27	4.16	5.60	5.86	5.59	4.81	3.11	1.57	0.67	0.51		
	RadStd		0.16	0.22	0.46	0.37	0.43	0.53	0.44	0.48	0.35	0.26	0.13	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.70	N/A	N/A	N/A	N/A	N/A	N/A	-160	N/A	+42
(47) Regional (7 neighbors)		+0.73	N/A	N/A	N/A	N/A	N/A	-146	-224	+107	+39

Nomenclature: See separate page